AI BlogNest

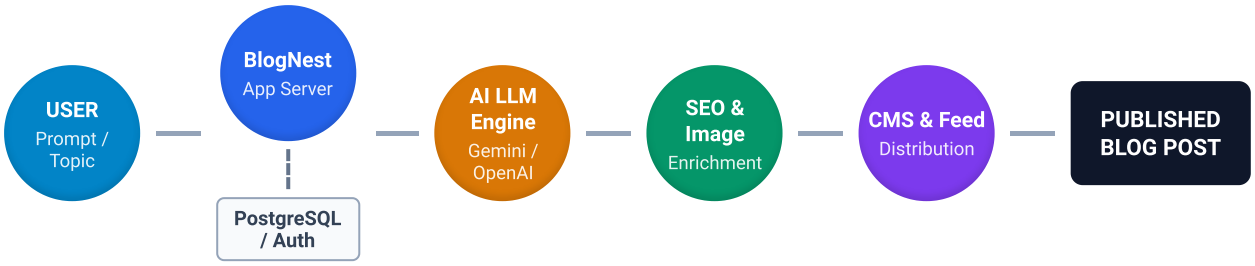
PROJECT DESIGN PHASE-II — DATA FLOW DIAGRAMS & COMPREHENSIVE USER STORIES

Date	29 September 2026	**Project ID**	SWTID-2026-2363
Project Name	AI BlogNest	**Max Marks**	4 Marks

Data Flow Diagram (DFD) Definition: A visual representation of the information flow within AI BlogNest. It depicts system inputs, outputs, processes, and data stores to graphically communicate software requirements.

1. Simplified System Flow Diagram

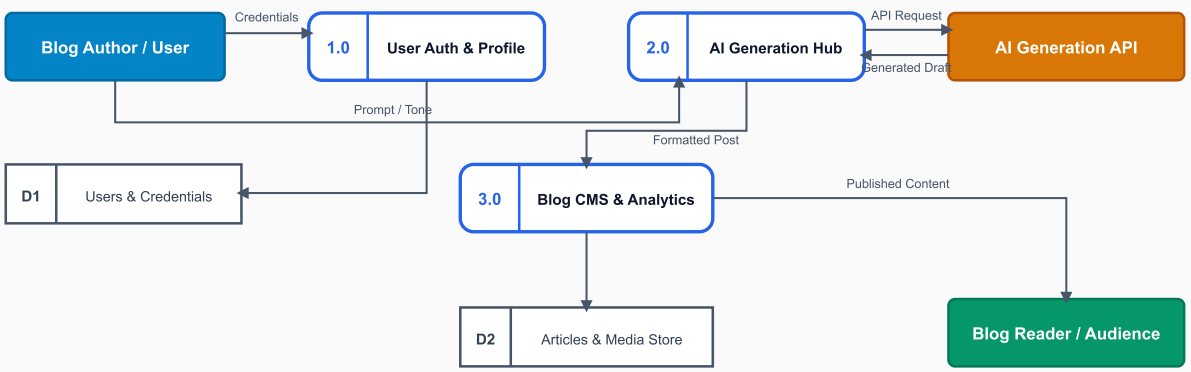
AI BLOG ENGINE ARCHITECTURE PIPELINE



Flow Execution Steps:

1. User configures preferences, inputs blog topic/keywords, and submits creation request.
2. BlogNest Backend authenticates user tokens and constructs structured prompt payloads.
3. AI Engine generates long-form blog content, automatic summaries, and key metadata tags.
4. SEO & Image Enrichment module extracts keywords, runs readability checks, and fetches hero visual.
5. Finalized article is stored in Database and rendered dynamically on modern web/mobile feeds.

2. Industry Standard DFD Level 0 (Context Diagram)



3. Product User Stories Matrix

Functional Requirements & Agile Backlog

User Role	Epic / Feature	ID	User Story / Task Description	Acceptance Criteria	Priority	Sprint
Customer (Mobile User)	Registration & Onboarding	**USN-1**	As a mobile user, I can register using email and password or social OAuth so I can start generating blogs.	Account created; Dashboard accessible immediately.	High	S-1
Customer (Mobile User)	Mobile Feed & Reading	**USN-2**	As a mobile user, I can view trending AI blogs in a responsive dark/light mode feed.	Posts render under 1.5s with clean typography.	High	S-1
Content Creator (Web User)	AI Blog Generator	**USN-3**	As a creator, I can enter a topic, select tone, target length, and keywords to generate a complete draft.	Draft is generated within 10s with markdown support.	High	S-1
Content Creator (Web User)	Rich Editor & SEO	**USN-4**	As a creator, I can edit the AI draft, auto-generate meta descriptions, and score SEO optimizations.	Live SEO score updates on keyword density changes.	Med	S-2
Content Creator (Web User)	Multi-Channel Export	**USN-5**	As a creator, I can export generated posts as PDF, Markdown, or publish directly to WordPress/Medium.	One-click publish returns valid live post link.	Low	S-3
Platform Administrator	User Management	**USN-6**	As an admin, I can monitor user activity, manage token usage limits, and grant premium access tiers.	Admin panel displays user token usage analytics.	High	S-1
Platform Administrator	AI Model Config	**USN-7**	As an admin, I can update API keys and switch fallback LLM providers to ensure zero platform downtime.	Seamless fallback executed on API failover test.	Med	S-2